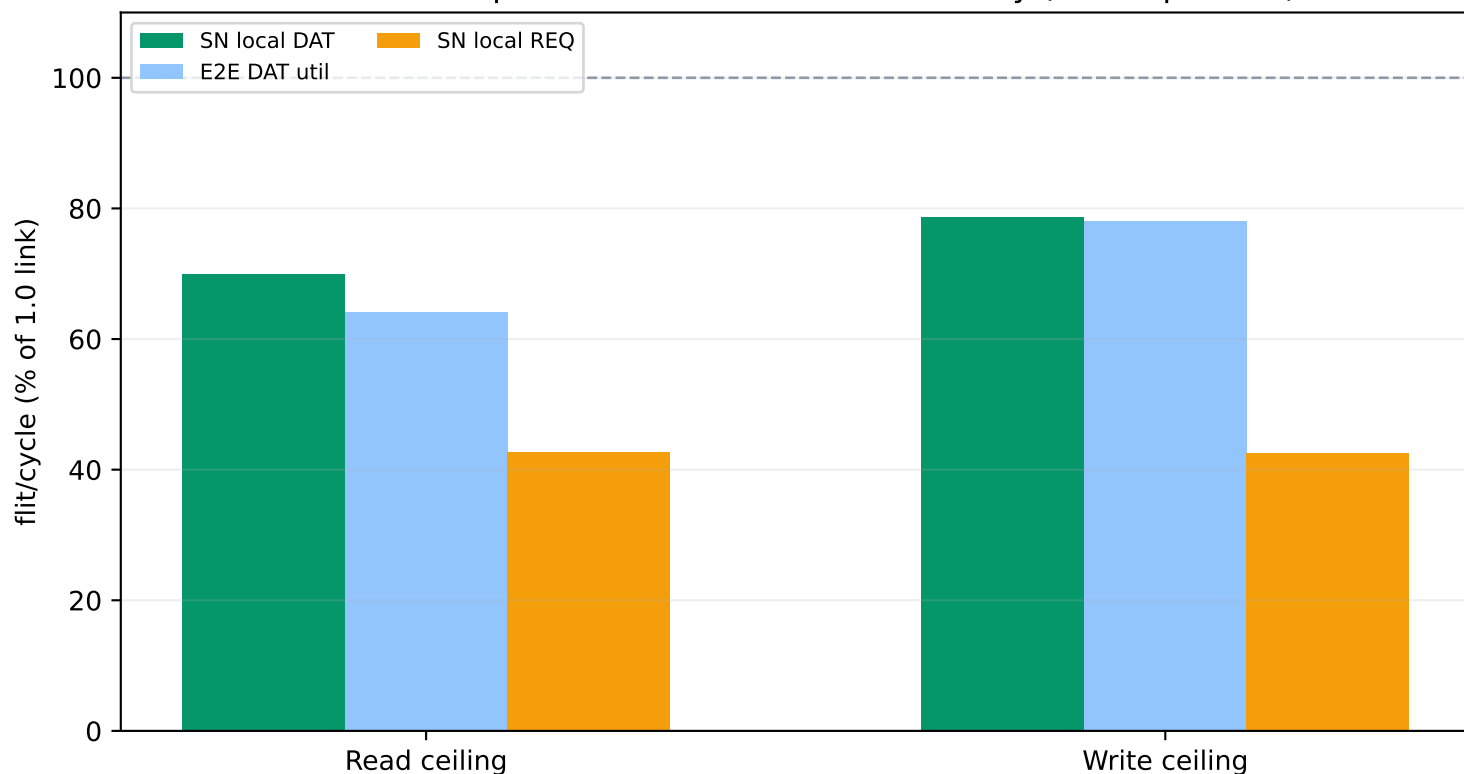


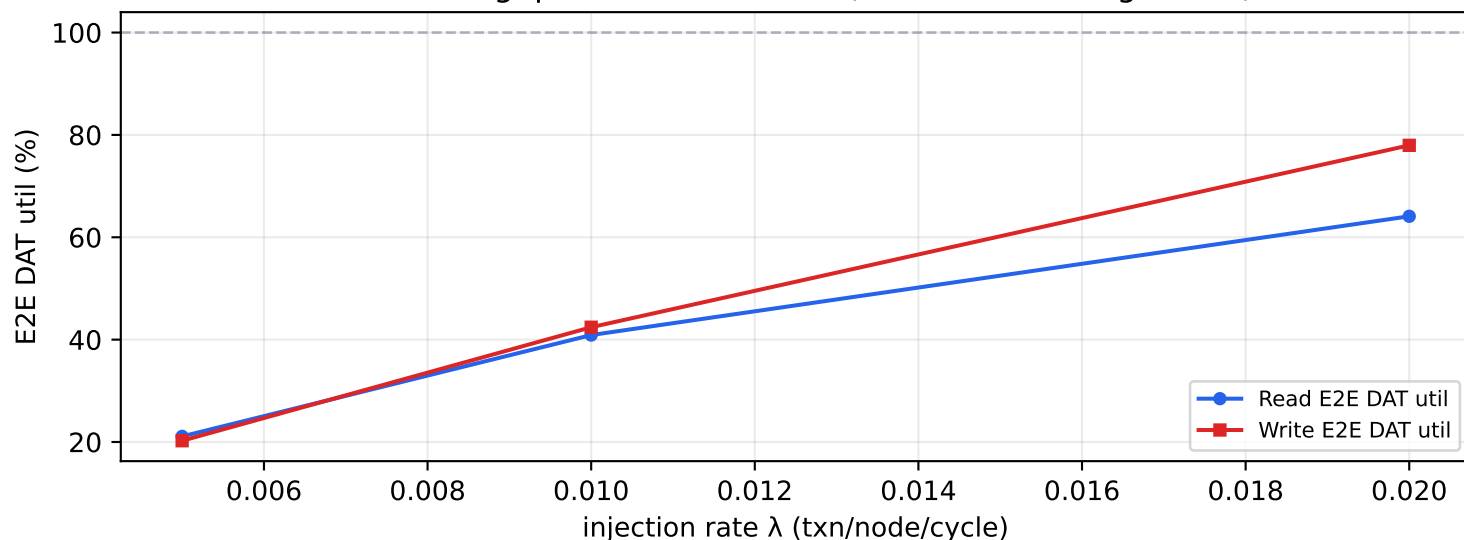
ZCN SN Read/Write Utilisation

vc_buf_size=4 flits/VC, DATA_FLITS=2, XY routing, link_latency=2

SN terminal peaks vs end-to-end DAT delivery (best λ per mix)



E2E throughput vs offered load (read/write ceiling mixes)



Path	λ^*	SN DAT peak	SN DAT avg	E2E DAT	SN REQ peak	Gap (SN-E2E)	State
Read	0.02	69.9%	64.0%	64.1%	42.7%	+5.8%	UNSTBL
Write	0.02	78.7%	78.0%	78.0%	42.5%	+0.7%	UNSTBL

Config: 6x7 mesh CHI, 4 SN nodes (168-171), vc_buf=4 flits/VC, 2 VCs, DATA_FLITS=2 (32B/packet on wire).
 Read ceiling: 100% ReadShared DMT miss $\rightarrow$ CompData SN $\rightarrow$ RN. Write ceiling: 100% WriteBack + L3EvictToSN.
 SN local DAT read = max sent_flits@SN (inject); write = max accepted_flits@SN (eject). E2E DAT = sweep_sn_throughput accepted-rate metric.
 Gap>0 on read $\Rightarrow$ mesh fan-out limits delivery below SN inject capacity. Gap=0 on write $\Rightarrow$ SN terminal link is the bottleneck.
 Write/read SN DAT peak ratio: 1.13x (79% vs 70%).